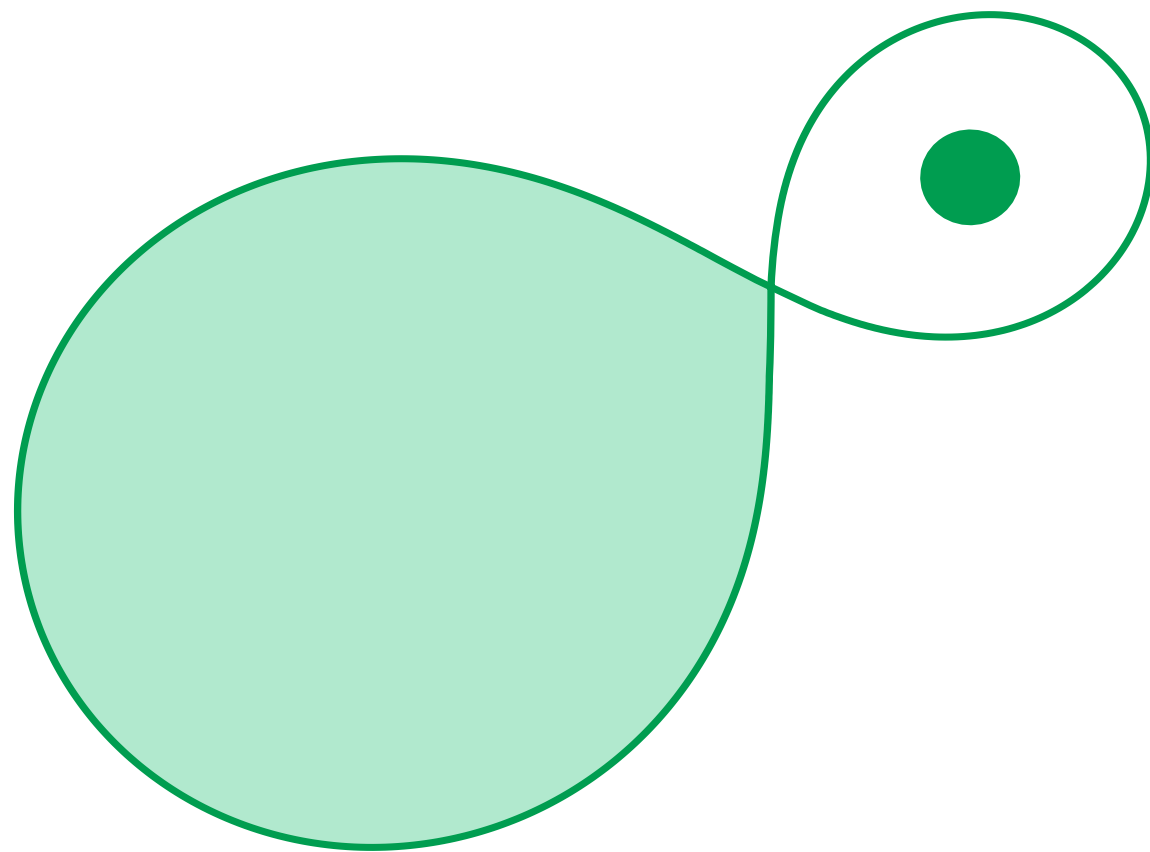


Forming barium stars through stable RLOF mass transfer from TPAGB-stars

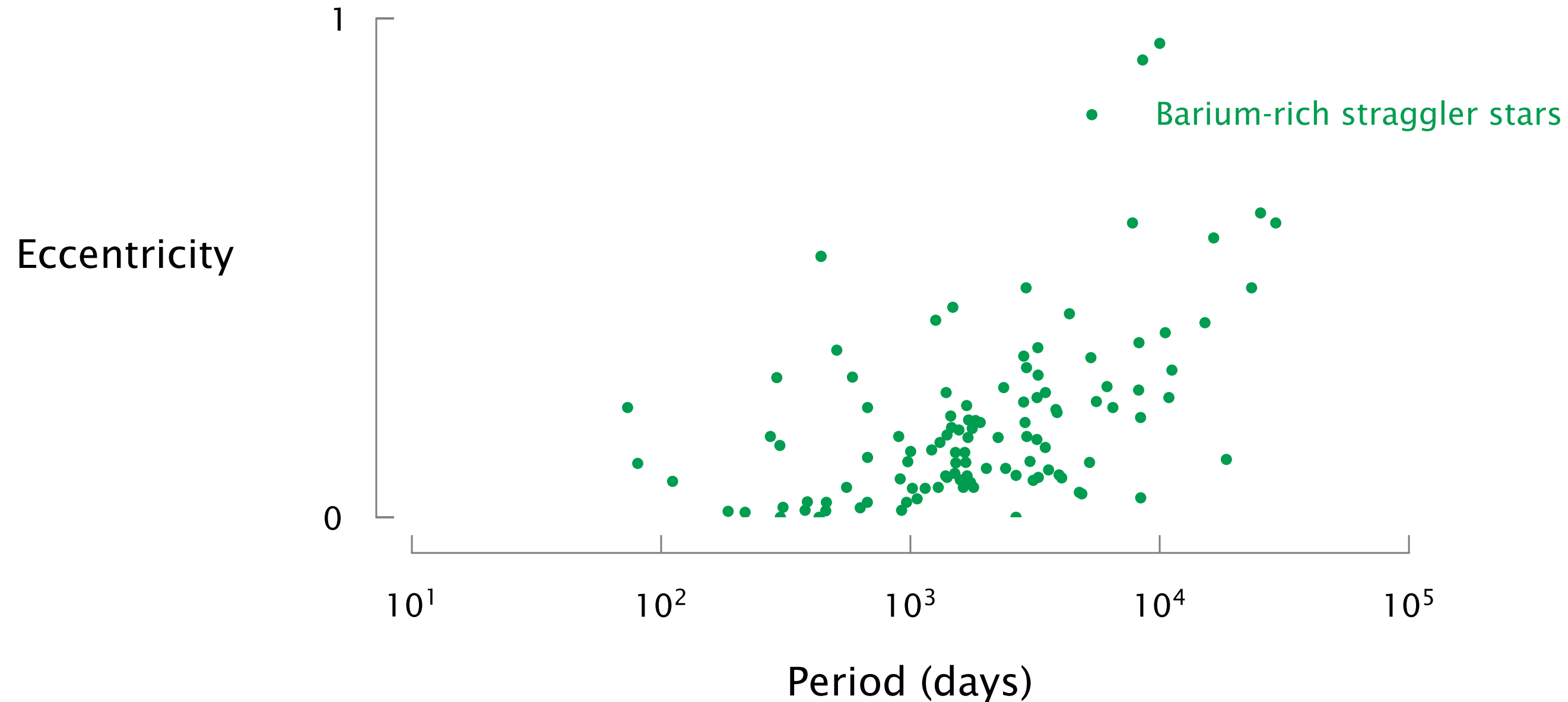


Koen Essers

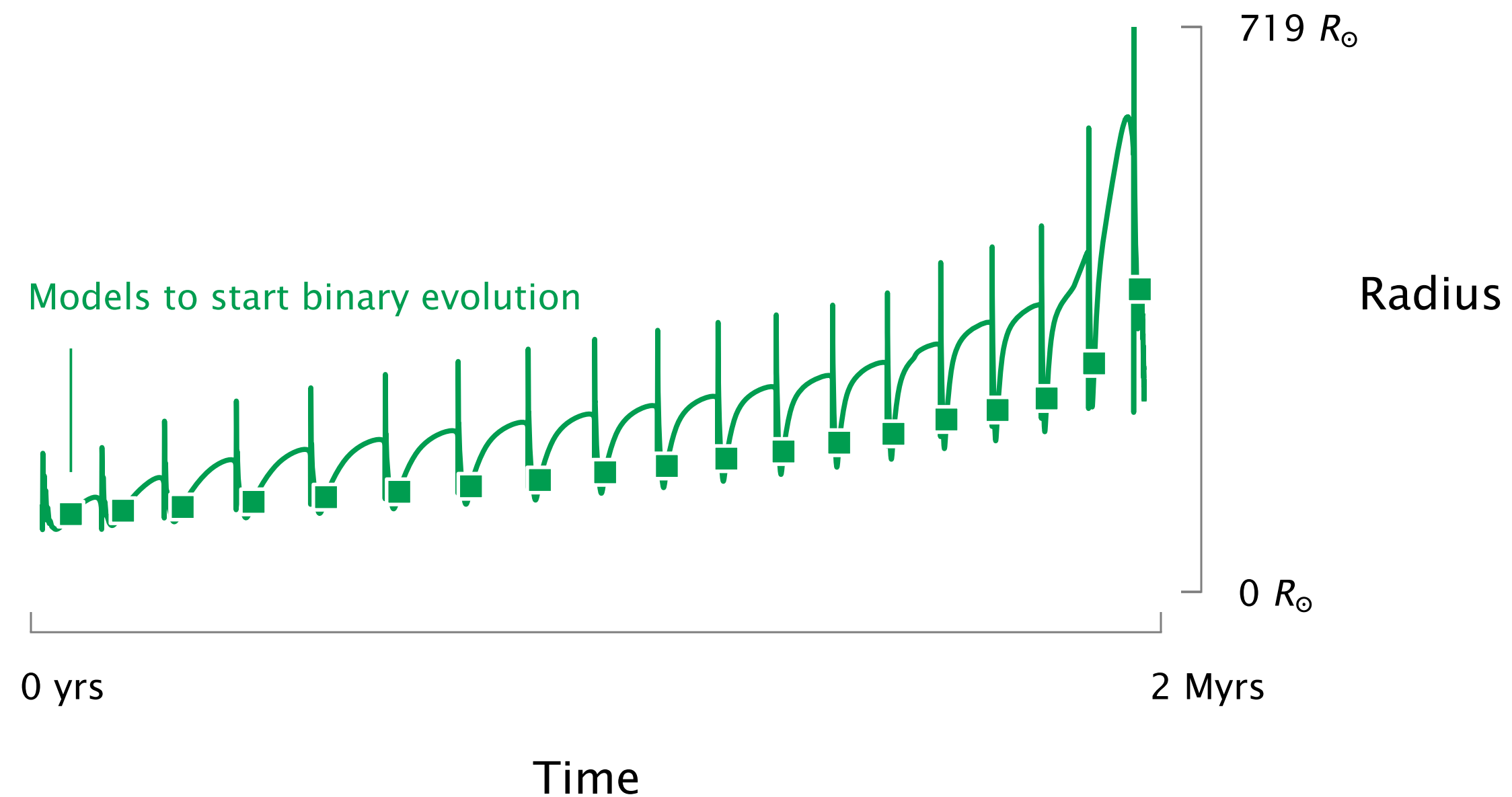
Radboud University

10th of June 2026

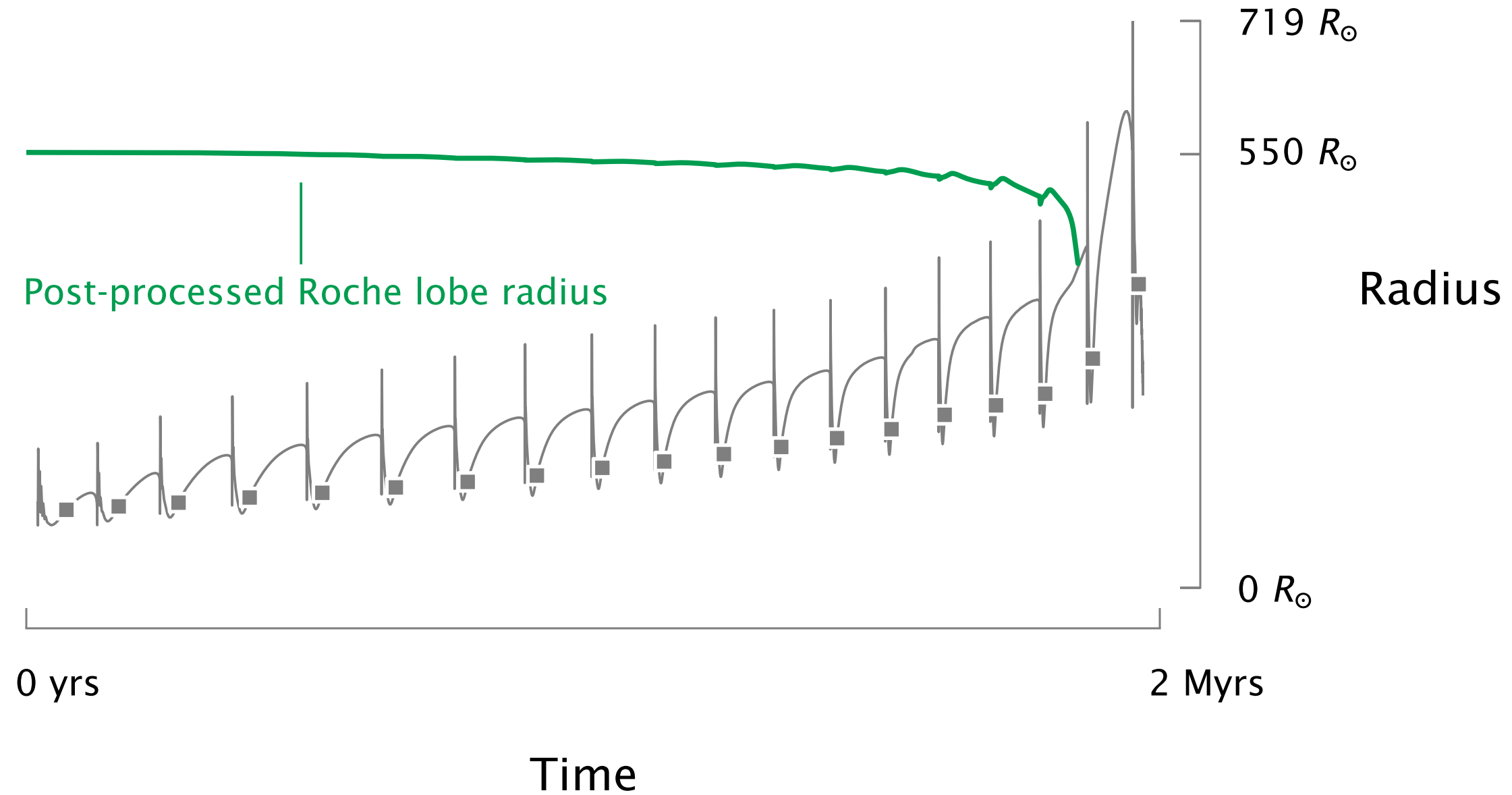
Ba stars have periods incompatible with wind mass transfer or common envelopes



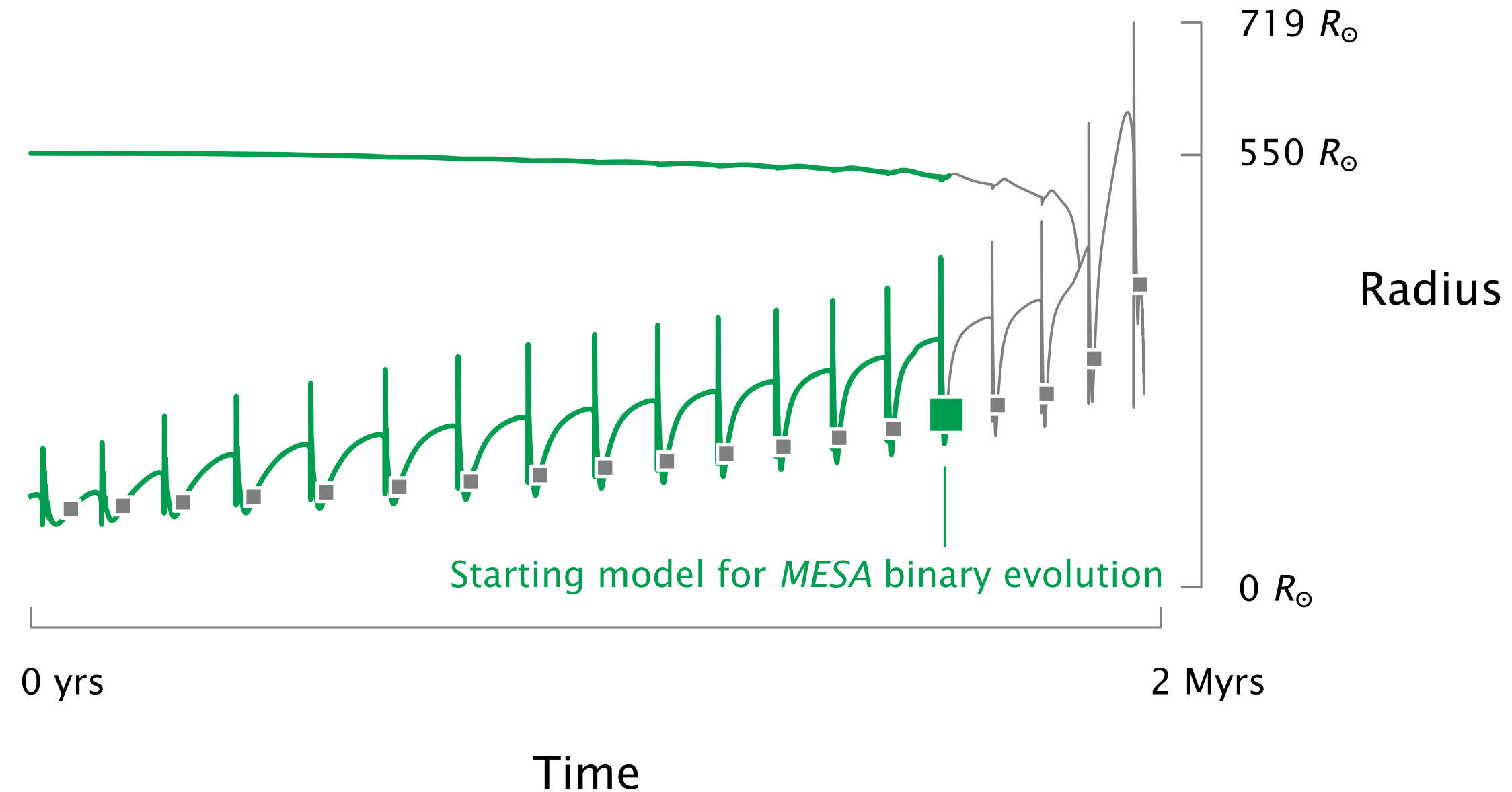
I simulate detailed TP-AGB stars with *MESA*
and I save models to use as binary input



I predict how the binary evolves using single star models and a post-processing step



I determine a *MESA* starting model and
infer the rotation, separation and mass-ratio



Tides, wind mass-transfer and stellar evolution drive systems into RLOF

